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MEASURES OF CONCENTRATION

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This paper discusses measures of industry concentration—the concentration ratio, the Herfindahl index and a new measure developed by the authors—both analytically and empirically. The analytical analysis consists of developing a set of properties which we argue all measures of concentration should possess. Although the concentration ratio is shown to be deficient on analytical grounds it appears to yield estimates of concentration not too different from the Herfindahl index and the measure devised by the authors.

INTRODUCTION

MANY attempts have been made to define and estimate a measure of concentration of firms within an industry that has a sound theoretical underpinning.¹ To date, however, a measure that is well grounded in economic theory has eluded the researchers. This paper suggests that traditional economic theory does not yield operational insights as to how to measure concentration and tries to (1) set forth some desirable properties of concentration measures, (2) examine the existing measures in the light of these properties, (3) develop and estimate a new measure of concentration, and (4) empirically compare and analyze the proposed measure with other existing measures. The main concern of this paper then is with measures of industry concentration as distinct from “overall” concentration that focuses on the largest 100 or 200 firms in the economy.

We assume that firms or establishments can be classified into meaningful industry groups and do not attempt to resolve the thorny questions concerning the size dimension—employment, assets, sales, value added—that should be used in computing a measure of concentration. Moreover we suspect that a universal answer to this question does not exist and therefore the concept of industry or size should be adapted to the questions or problems being analyzed.

Measures of concentration are used to answer cross sectional questions concerning the variation in concentration among industries in a given time period, time series questions concerning changes in concentration in a particular industry over time and questions that incorporate both time series and cross sectional information.

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¹ In this regard see [8, p. 55] where the Herfindahl index is derived as a “measure of the likelihood of effective collusion” and two circulating but as yet unpublished manuscripts, “An Informational Approach to the Measurement of Concentration,” Henri Theil, Myron Scholer, Pedro Uribe-Report 6512, Center for Mathematical Studies in Business and Economics, University of Chicago and “Concentration Ratios, The Degree of Monopoly and The Share of the 250 Largest Manufacturing Firms,” T. R. Saving, Michigan State University. Theil et al develop a measure based on informational theory which uses the entropy of the relative shares as an inverse concentration measure. Saving attempts to derive a relationship between the Lerner measure of monopoly and the traditional concentration ratio CR .

Cross sectional comparisons require a common measuring device across all industries and this requires a method of combining the P_i 's—the points on the concentration curve. P_i is the percentage of a size measure, such as sales, controlled by the i th firm and a concentration curve is a cumulative plot of the P_i 's with the firms ranked by size on the horizontal axis and the P_i 's on the vertical axis. The problem is to find a method of combining the P_i 's as concentration curves of different industries may intersect when plotted on the same graph. The intersection of the concentration curves means that it is easy to find weighting schemes which respectively would make a particular industry less concentrated, equally as concentrated or more concentrated than some other industry. Selecting a set of weights cannot be avoided, however, as the unweighted sum of the P_i 's would always be equal to one regardless of the number of firms in a particular industry. Economic theory, however, provides little guidance, other than the notion that many sellers mean low concentration and few sellers mean high concentration, as to how the P_i 's should be combined. The fact that the number of firms in SIC four digit classes ranges from 5 (SIC 3332) to 15,731 (SIC 2421) emphasizes the vagueness in concepts like "many" and "few" [9, p. 19, 30].

The method of combining the P_i 's is not unimportant for analyzing changes in concentration in one industry over time, but it is much less crucial since the intra-variation in the structure of a particular industry is much less than the structural inter-variation among industries. Needless to say as we extend the time period the intra-variation could become quite severe.

1. SOME PROPERTIES OF MEASURES OF CONCENTRATION

We shall assert that a measure of concentration ought to have several properties.²

(1) Concentration is a one dimensional measure. Given two industries A and B (for any measure of concentration), A is either more concentrated than B , less concentrated than B or the two are equally concentrated. A measure of concentration should in other words be unambiguous. Measures of more than one dimension, such as that proposed by Fellner, reflect we think the difficulties of developing a single parameter measure rather than a multi-dimensional concept of concentration and are generally ambiguous [2, p. 115].³

(2) Concentration in an industry is independent of the size of the whole; it is some function of the relative shares of firms in the industry, that is, some function of all the P_i 's.⁴

Adelman has suggested that the total number of firms in an industry is "usually unknowable with any precision" which might lessen the operability of measures dependent on all the firms in an industry [1, p. 69]. We submit, however, that this criticism must also apply to any measure based on relative

² Compare Fisher's criteria for index numbers [3] and Niehans [5]. The properties enunciated above are somewhat similar to Niehans though derived independently of his article. The writers are grateful to a reader for bringing the Niehans article to their attention.

³ Fellner proposed a measure such as 20-3-52 (s) which would mean that "the largest firm has a share of 20%, that the total number of firms with a share of more than 10% is three, that the joint share of these three firms 52% and that very small firms account for more than 10% of the total" [1, p. 115].

⁴ Stigler states in [7, p. 7] that . . . "I would expect to find also that the absolute number of firms . . . were important in influencing the industry's behavior."

shares as we cannot compute a relative value without knowing the total value.

(3) A measure of concentration should be affected by a change in any P_i , with concentration increasing if (for a small change) there is a shift from a lower ranked firm to a higher ranked firm, and vice versa. More formally, define $\delta C / \delta P_i^j$ as the rate of change in a measure of concentration C when sales are shifted from the i th largest firms to the j th largest.⁵ If $P_i > P_j$ (i.e. $j > i$) the rate of change should be negative; concentration should decrease. If $P_i < P_j$ (i.e. $i > j$) the rate of change should be positive; concentration should increase.

(4) If industry A had K times the number of firms in industry B , $K > 1$, and the P_i 's in A are distributed such that corresponding to each P_i in B there are K firms of size P_i/K then the measure of concentration for A should be $1/K$ times the measure for B . This rule holds also for changes in a given industry over time. Thus if each firm in a given industry is divided into two firms of equal size, the effect on a measure of concentration should be to reduce it by one-half. This property is very necessary if (with no specific guidelines as to an appropriate weighting scheme) we are to have confidence in the cardinal properties of the measure.⁶

(5) When an industry is divided into N equal sized firms, a measure of concentration should be a decreasing function of N . This is implicit in properties (3), (4) and (5) discussed above, but it is worth noting since it highlights the difference between concentration and measures of inequality typified by the Lorenz curve. The Gini coefficient of the Lorenz curve would not distinguish between a single firm industry and one with a thousand equal sized firms. In addition this property incorporates the only specific guide that can be gleaned directly from economic theory—many firms mean less concentration, fewer firms mean more concentration.

(6) A measure of concentration should have a range of 0 to 1. While this property is not strictly necessary, it makes the measure easier to use, and it is always possible to perform a transformation on a measure that will give it this range without changing its order properties.

There are many measures of concentration consistent with the above properties and there is nothing inherent in the properties which permits us to choose between measures. This, however, is not so much a fault of our properties as it is a basic problem of measures of concentration or more generally index numbers. The reference to index numbers is we think, apt, and helps us to get to the root of the problem which is the selection of an appropriate set of weights with which to combine the respective P_i 's. The weighting problem here is similar to the weighting problems with any index number and as with index numbers there are no *a priori* reasons free of crucial assumptions which will allow us to say that a particular weighting scheme is best. The properties set forth above cannot uniquely determine the best measure of concentration but they do serve to delineate measures that are undesirable for theoretical reasons.

⁵ Let the change in concentration be D , then the

$$\frac{\delta c}{\delta P_i^j} = \frac{g(P_1, P_2 \dots P_i - D \dots P_2 + D \dots P_n) - g(P_1, P_2 \dots P_i, P_j \dots P_n)}{D}$$

⁶ This property does not imply anything about the relation between concentration and other economic variables.

2. APPRAISAL OF EXISTING MEASURES

The existing measures that will be appraised in the light of the above properties are the concentration ratio— CR , and the Herfindahl-Hirschman index— HH . Other measures not discussed because their limitations are similar to the CR are the area under the concentration curve of the largest L firms—generally 4, 8 and 20 and the number of firms that account for a certain percentage—generally 80 of a given industry's size.

The CR is the fraction of industry size held by the largest L firms— $\sum_{i=1}^L P_i$ —where L is usually 4, 8 or 20. The CR is, in other words, a point on the concentration curve and is thus a one dimensional measure that has a range of 0 to 1. When the industry consists of N equal sized firms CR is L/N —appropriately a decreasing function of N . CR 's then satisfy properties 1, 5 and 6. However, $\delta CR/\delta P_i^j = 1$ for $i < L$ and $j > L$; $= -1$ for $i > L$ and $j < L$; and 0 elsewhere. The fact that the $\delta CR/\delta P_i^j = 0$ in some cases violates property 3 and is formal recognition of the fact that the CR implicitly treats firms of rank less than L as not affecting the measure of concentration. It follows from the above that the CR fails also to satisfy property 2 since it does not include all the P_i 's.

Property 4 is also violated as it is only under very special circumstances such as where all firms are of equal size that dividing each firm into two equal sized firms would decrease the measure by one half.

Using only the L largest P_i 's and then arbitrarily assigning each of these firms the same weight are the main reasons for the failure of the CR to satisfy properties 2, 3 and 4.

The HH index— $\sum P_i^2$ avoids the basic shortcomings of the concentration ratio as it utilizes all the P_i 's. Indeed it satisfies all of the properties discussed above.

Properties 1, 2 and 6 are obviously properties of this index since it is by definition a measure that is one dimensional, utilizes all the P_i 's and varies between zero and one. Property 3 is satisfied since the $\delta HH/\delta P_i^j = 2(P_j - P_i)$. The $\sum_{i=1}^N K(P_i/K)^2$, which expresses the requirements of the fourth property, reduces to $1/K \sum (P_i)^2$ and when the industry consists of N equal sized firms the $\sum P_i^2 = NP_i^2 = N(1/N)^2 = 1/N$ thus making HH consistent with properties 4 and 5. The HH index by our criteria emerges as superior to the CR .

3. A NEW MEASURE OF CONCENTRATION

Although the HH index satisfies all of our properties it should not, as we pointed out above, be regarded as ideal since many measures are consistent with the above properties. The HH index weights each firm by its relative share (weights increase from the smallest firm to the very largest) which implies that the relative sizes of firms are more important than the absolute number of firms in determining concentration in an industry. One can, however, argue that the absolute numbers of firms in an industry (since they indicate in part the nature of entry conditions—many firms easy entry, few firms entry blockaded) should be heavily stressed in a measure of concentration.

A measure which weighted each firm by its rank—the i th largest firm receiving weight i —is one method to ensure that the absolute number of firms is emphasized.

The measure $\sum iP_i$ immediately suggests itself. Moreover, if sales are an appropriate measure of size then P_i may be regarded as the probability that a firm of rank i will obtain a random dollar of sales and $\sum iP_i$ is then the expected value of that rank. This measure has a range of 1 to ∞ with concentration decreasing in the direction of infinity. Appropriately the greater the expected value of the rank of a firm gaining a random dollar of sales the less the concentration.

Its reciprocal $1/\sum iP_i$ has the desired range and is comparable to HH and CR . $1/\sum iP_i$ does not satisfy property 4, however, and we accordingly transform it to $1/(2\sum iP_i) - 1$ which satisfies all six properties and has the same order properties as $1/\sum iP_i$.⁷ This index— TH —weights each firm by its rank, thus giving emphasis to the absolute number of firms.

It meets the requirements of properties 1, 2 and 6 by definition. Property 3 is satisfied since $\delta TH/\delta P_i^j = i - j/2(TH)^2$; it can be shown to be consistent with the fourth property and the fifth property holds since $TH = 1/N$ when the industry consists of N equal sized firms.

4. EMPIRICAL ANALYSIS

CR , HH and TH differ analytically because of their different weighting patterns. It is possible, however, as has been argued elsewhere for CR and HH that these differences are insignificant empirically [4, pp. 63–69]. Differences here refer to the ranking of industries by degree of concentration and to the relative changes in concentration over time; not to actual differences which would of course exist since the measures are not analytically or computationally the same.

The rank correlation coefficients between the various measures of concentration for 446 SIC four digit industries, 1958 data, are $CR-HH$.995, $CR-TH$.904 and $TH-HH$.933.⁸ Using HH and CR measures developed by Nelson for 76 SIC four digit industries, 1957 data, we found a rank correlation of .959 between CR and HH [4, p. 25ff]. Nelson's estimate of HH should be quite accurate since he used individual firm data up to the fiftieth largest firm in each industry. These very high rank correlations suggest that with respect to ordinal qualities there is scarcely any difference between the three measures.

Despite the high rank correlation coefficients between the measures a careful examination of the rankings revealed several instances where the difference in the rankings was substantial. In 23 of the 446 industries the difference in the rankings between TH and HH or TH and CR was greater than 90.

In general the TH measure yielded somewhat different rankings (less concentrated relative to those of CR and HH) for industries with several hundred firms where a few firms—under 10—accounted for 30% or more of the total

⁷ This transformation is necessary since if each existing firm is split into equal sized firms [Property 4] $\sum^N P_i = C$ would change from C to $2C - \frac{1}{2}$. Subtracting $\frac{1}{2}$ from $\sum^N iP_i$ would of course satisfy property 4. The range of the measure now becomes $-\frac{1}{2}$ to ∞ , however, so we multiply by 2 to obtain $2\sum iP_i - 1$ and restore the range to 1 to ∞ . The reciprocal $\frac{1}{2}(\sum iP_i) - 1$ is our measure of concentration.

⁸ HH and TH were computed from Table 2 (Value of Shipments) of the Concentration ratios in Manufacturing Industry 1958, Subcommittee on Antitrust and Monopoly of the Committee on the Judiciary, Washington: 1963. This Table provides the number of firms and concentration ratios for 4, 8, 20, and 50 firms in each industry. In computing HH and TH we assume that firms in each class 0–4, 5–8, 9–20, 21–50, 51– N —the number of firms in each industry—are of equal size. This approximation provides minimum estimates of concentration for HH and TH .

size of the industry and (more concentrated) where the largest few firms accounted for a small amount of the total size of the industry and the total number of firms in the industry was small.

The normal simple correlation coefficients between the measures was also very large, $CR-HH$.976, $CR-TH$.883 and $HH-TH$.947.

The actual range of the measures were CR .03 to 1.0, HH .0012 to .25 and TH .00008 to .25 with $CR > HH > TH$, in general. TH on the other hand had the greatest dispersion as measured by the coefficient of variation—127.01 as compared with 90.29 for HH and 57.28 for CR . This greater dispersion suggests that *ceteris paribus* TH is a more sensitive measure of concentration.

Nelson estimated CR 's and HH 's for 1947, 1954–1957 [4, p. 111ff]. In order to determine if the two measures yielded, on the average, the same relative changes in concentration we correlated the ratios CR_t/CR_{t-1} and HH_t/HH_{t-1} over all industries with 1947 treated as $t-1$ with reference to 1954. If CR and HH yielded the same rates of change we would expect a simple correlation = 1. The actual " r " was .846 which though large is significantly different from one since a confidence interval of $\pm 3\sigma$'s with 312 observations yields a range of .789 to .887 via Fisher's z transformation. It should be noted, however, that the mean values of the ratios of each measure are at the 95% level not significantly different from each other— CR 's mean ratio = 1.00 and that for HH = 1.03.

Nelson does not report the number of companies in each industry for 1955–1957 and computes separate concentration ratios for only the largest 4, 8 and 20 firms thus preventing the computation of TH . Given, however, that TH , as discussed above, differs from the other two measures more than they do from themselves we suspect that comparing rates of change between TH and any of the other two measures would show a wider divergence than the comparison between HH and CR .

CONCLUSION

We have set forth a few desirable properties of measures of concentration and shown that with respect to these properties there are better measures than the popular concentration ratio. $HH = \sum P_i^2$ and $TH = 1/(2 \sum i P_i) - 1$ satisfy all our properties despite obvious differences in the method used to assign weights to the various P_i 's. It is our belief that no *best* measure of concentration exists and that in general the measure should be suited to the use.

HH and TH share with CR the property that there is more than one distribution of firms which can generate a given value of the index. We do not think this is a serious shortcoming, however, because, as pointed out by a reviewer, this is typical of measures that use an "averaging procedure." Moreover since concentration refers both to inequality and the number of firms in the industry we should expect that under some situations the trade off between numbers and inequality would be such as to allow industries with different size distributions to be viewed as equally concentrated.

Our empirical analysis revealed that despite its analytic shortcomings the concentration ratio yields cross sectional rankings of concentration among industries that are generally similar to those of HH and TH . Indeed our findings

imply that if HH or TH is the correct measure of concentration then the concentration ratio is certainly a good proxy. The concentration ratio may not be a good measure of monopoly but as a measure of concentration it seems to hold up where it counts, that is in the empirical analysis. It should be stressed, however, that data limitations prevented us from developing estimates that would exploit all the subtleties of HH and TH .

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